# Exp:4-Rutherford Scattering Experiment

November 2024

## 1 Overview

In this experiment we acquired data for different scattering angles on either sides of the incident beam direction. The fixed detector is called the RBS and the movable detector is called the PRD. Here we recorded the dead time values for both detectors and when taking the number of counts we conducted the dead time correction. Initially we did the energy calibration and the residual energies seem to be extremely small and then the bin number of the copper and silver peaks were found. These values seem to match the with the obtained data proving that the energy calibration done is accurate. Afterwards the next task was to obtain the composition of the elements. It was observed that the composition of gold as 79.28, silver as 10.99 and copper as 9.72. The next task was to plot the differential cross-section vs the angular distribution this was done for the low energy value of 2.5 MeV.
When analyzing the peaks it was observed that at low energy values the peaks for copper and silver aren't prominent as when the energy is at a higher value.

In Figure 1 it is seen that the gold, copper and silver peaks are superimposed into one peak. Therefore,

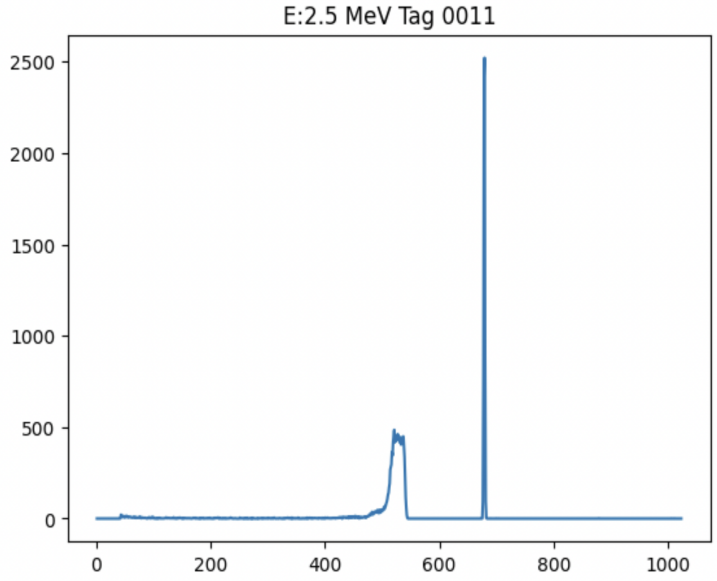



Figure 1: Energy: 2.5 MeV

obtaining the peaks separately was not ideal. In Figure 2 it can be seen that the gold, silver and copper peaks are visible separately and therefore its easier to analyze.

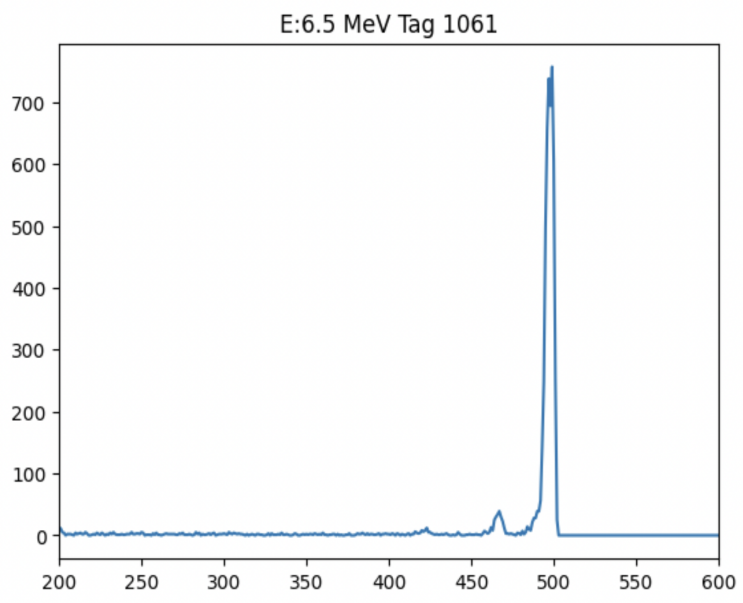



Figure 2: Energy: 6.5 MeV

## 2 Energy Calibration

The k value was calculated using the given equation:

$$k = \left( \frac{M_0 \cos(\theta) + \sqrt{M_i^2 - M_0^2 \sin^2(\theta)}}{M_0 + M_i} \right)^2$$

$M_0$ is the atomic mass of helium and $M_i$ is the atomic mass of the corresponding elements: Gold, Copper, Silver.
Here we are taking into account the Energy loss in the foil. $E_{avg}$ is calculated using $E_0$ and $E_f$ where $E_f$ is the alpha energy after passing though the gold foil.

$$E_{avg} = \frac{E_0 + E_f}{2}$$

Then,

$$E' = E_{avg} k$$

The energy calibration was done to different energy values: 3.5 MeV, 4.5 MeV and 6.5 MeV. The forward angle is 55 deg and the backward angle is 155 deg.
The fit function is: y = (0.0118 ± 5.7840e-5)x + (0.1307 ± 2.2743e-2)

Using the energy calibration we obtained the corresponding bin counts for copper and silver.

Table 1: Residual Energy values

| Energy calculated (MeV) | Energy calibrated (MeV) | Residual (MeV) |
|---|---|---|
| 3.45 | 3.45 | 0.0012 |
| 4.44 | 4.43 | -0.0038 |
| 6.40 | 6.39 | -0.0137 |
| 3.24 | 3.23 | -0.0130 |
| 4.16 | 4.18 | 0.0185 |
| 6.02 | 6.03 | 0.0106 |

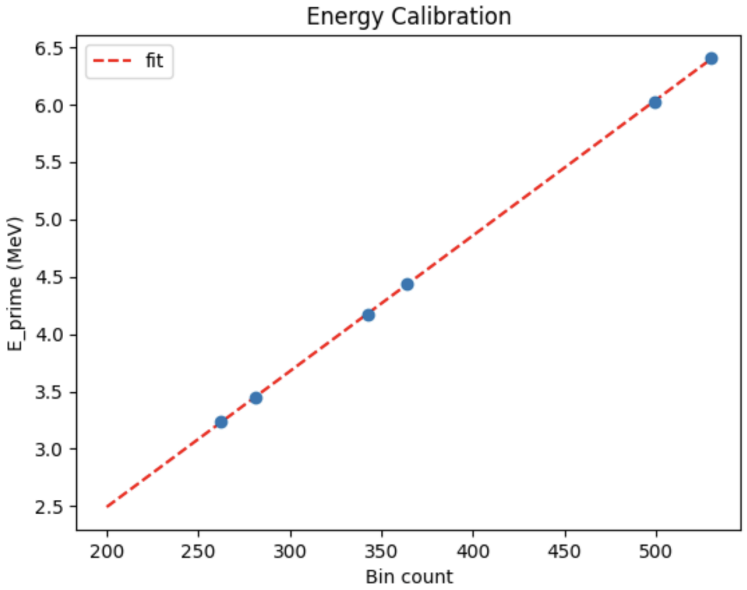



Figure 3: Energy Calibration

Table 2: Bin counts and Energy for Silver and Copper

| Energy: Cu | Bin counts: Cu | Energy: Ag | Bin counts: Ag |
|---|---|---|---|
| 3.3459 | 272.1326 | 3.4084 | 277.4201 |
| 4.3020 | 353.0475 | 4.3822 | 359.8459 |
| 6.2140 | 514.8771 | 6.3300 | 524.6974 |
| 2.7522 | 221.8841 | 3.0383 | 246.0935 |
| 3.5386 | 288.4421 | 3.9064 | 319.5688 |
| 5.1114 | 421.5580 | 5.6426 | 466.5192 |

# 3 Composition of foil in terms of gold, silver and copper:

The composition was calculated for the back scattering angle: 155 deg at the energy 6.5MeV. Theoretical differential cross-section is used in this context,

$$\frac{d\sigma}{d\Omega} = 1.296 \left(\frac{Z_0 Z_1}{E_0}\right)^2 \left(\frac{1}{\sin^4\left(\frac{\theta}{2}\right)} - 2\left(\frac{M_0}{M_1}\right)^2\right)$$

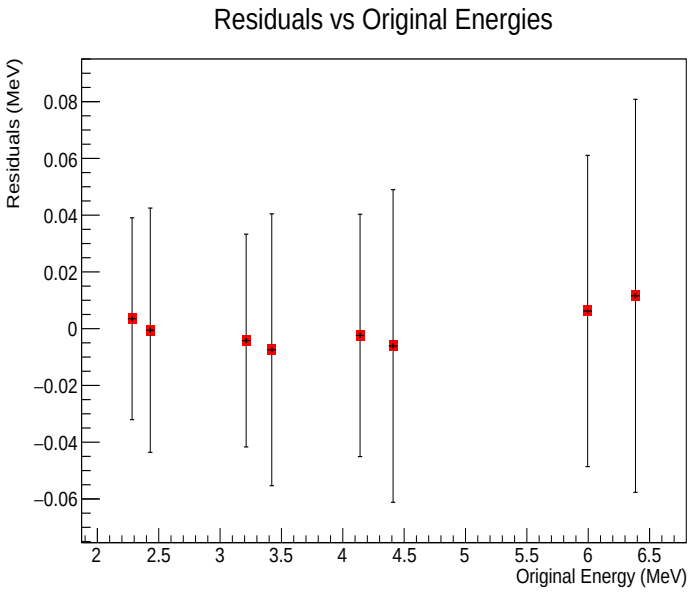



Figure 4: Energy Calibration

Then using the number of counts we were able to derive a formula for the composition percentage for the three elements.

$$R_{AgAu} = \frac{N_A g}{N_A u} \frac{d\sigma_{Au}}{d\sigma_{Ag}}$$

$$R_{cuAu} = \frac{N_c u}{N_A u} \frac{d\sigma_{Au}}{d\sigma_{cu}}$$

$$Gold = (\frac{1}{1 + R_{AgAu} + R_{cuAu}})$$

This method is similarly continued to obtain the composition of silver and copper as well.

Table 3: Counts and Theoretical cross-section

| **Element** | **Counts** | **Cross-section** |
|---|---|---|
| Gold | 4455.3 | 1091.55 |
| Silver | 218.74 | 385.68 |
| Copper | 73.1709 | 146.14 |

Gold(%) : 79.2819 ± 1.12075
Silver(%) : 10.9929 ± 0.650352
Copper(%) : 9.72527 ± 1.29578

# 4 Differential cross-section vs angular distribution

Here we are obtaining the relationship between the angular distribution and the cross-section. As observed in Figure 2, the theoretical value aligns with the experimental value.
Here the Energy is 2.5 MeV.

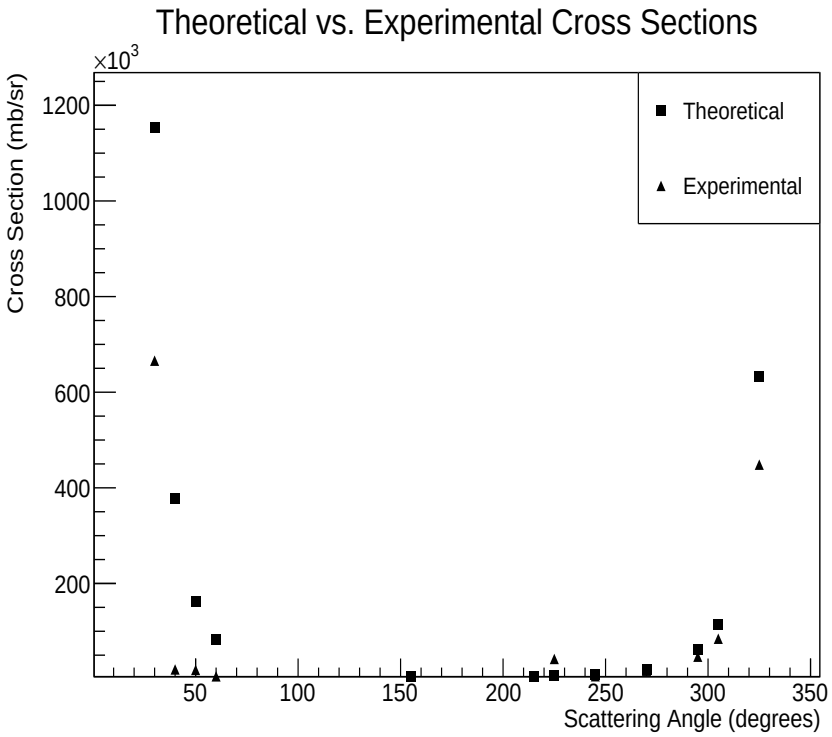



Figure 5: Angular Distribution

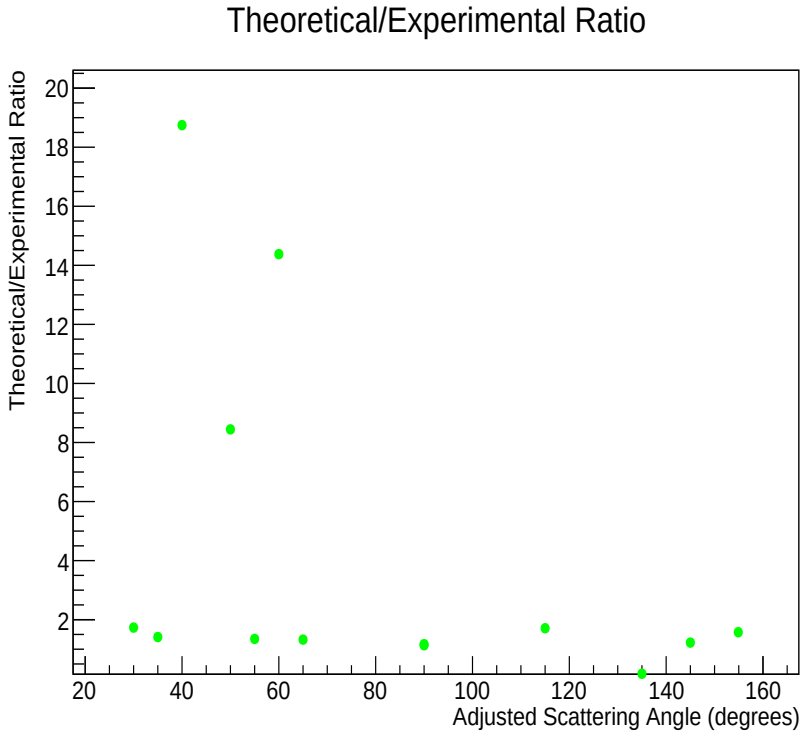



Figure 6: Ratio between theoretical and experimental cross-sections